Automatic Reconstruction of CAN Signal Definitions Using Linear Regression and Candidate Search

Math for AI Research Project

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Abstract

Controller Area Network (CAN) systems transmit large amounts of numerical data, but interpreting this traffic normally requires prior knowledge of the location, width, scaling, and meaning of each encoded signal. This project investigates whether hidden signal definitions can instead be reconstructed automatically from raw CAN traffic and synchronized measurements of the corresponding physical variables. The method assumes that numerical CAN signals follow a linear encoding of the form $y = ax + b$, where x is a raw integer extracted from a candidate location in a CAN payload, y is a known physical value, a is the signal scale, and b is the offset. Candidate signal definitions are generated systematically, fitted using least-squares regression, scored, and ranked without giving the discovery algorithm access to CAN database definitions, PGN-to-signal mappings, SPNs, scales, offsets, or other ground-truth information.

The approach was first developed using NMEA 2000 traffic and was subsequently generalized to SAE J1939. The largest experiment covered the complete supplied J1939 database: 2,043 message definitions containing 13,774 signals. Each signal was observed 200 times, producing 2,754,800 synchronized target observations, 408,600 logical messages, and 547,600 physical CAN frames. The full generic search evaluated 89,775,236 candidate regressions.

Across all 13,774 J1939 signals, the system recovered the correct PGN for 100% of signals, achieved 98.2576% Top-1 location accuracy, 99.1578% Top-10 location accuracy, 97.4808% strict full-encoding accuracy, and 98.7876% identifiability-adjusted full-encoding accuracy. Of the full signal population, 13,427 signals were classified as identifiable, 180 as structurally ambiguous, and 167 as incorrect. The results show that linear regression combined with systematic candidate enumeration can reconstruct the overwhelming majority of a large CAN signal database without direct access to the original signal map. They also demonstrate an important limitation: some properties, particularly signedness and declared field width, can be mathematically unidentifiable when different encodings produce exactly the same observable values over the complete valid range.

Keywords: CAN bus, J1939, NMEA 2000, linear regression, reverse engineering, signal inference, identifiability, machine learning

1. Introduction

Controller Area Network (CAN) is widely used to exchange data between embedded controllers. A raw CAN capture, however, provides only identifiers and payload bytes. Meaningful physical quantities such as speed, pressure, temperature, current, or engine speed are encoded somewhere within those payloads. A conventional decoder requires a predefined description of each signal: its message identifier or PGN, start bit, width, byte order, signedness, scaling factor, and offset.

This project asks whether those definitions can be inferred automatically when a second synchronized data source provides the physical values themselves. The inference algorithm is intentionally denied the original

database. It receives only raw physical CAN frames and a synchronized list of known values. It is not given CAN database definitions, PGN-to-signal mappings, SPNs, true bit positions, widths, scales, offsets, byte order, signedness, or semantic hints from names and units. Evaluation against the answer key takes place only after discovery has completed.

The underlying hypothesis is that many numerical CAN signals have a simple linear representation. If enough synchronized observations are available, the correct raw field should exhibit a near-perfect linear relationship with the known physical measurement, while unrelated fields should not.

$$y = ax + b$$

The project developed this idea in several stages. A synthetic NMEA 2000 simulator was first built from Canboat definitions. A protocol-neutral inference engine was then developed using aligned 8-, 16-, and 32-bit candidate fields before being extended to arbitrary bit positions and widths. The same inference architecture was later applied to a substantially larger J1939 database containing 13,774 signals. Finally, the search was expanded to the entire supplied J1939 library, including single-frame messages, BAM transport, and destination-specific RTS/CTS traffic.

Research question: How accurately can a protocol-independent linear inference method reconstruct an entire CAN signal database from raw traffic and synchronized physical values without access to the original signal definitions?

2. Mathematical Model

For a candidate raw signal x and known physical value y, the assumed relationship is linear. The regression therefore estimates only two parameters: the scale a and offset b.

$$y_i = a\,x_i + b$$

For n synchronized observations, the least-squares design matrix and parameter vector are:

$$X = [[x_1, 1], [x_2, 1], \ldots, [x_n, 1]]$$

$$\theta = [a, b]^T$$

$$\theta = \arg\min \|X\theta - y\|^2$$

Rather than training a neural network, the inference stage uses ordinary least-squares regression to estimate the scale and offset associated with every plausible raw CAN field. Each candidate is evaluated using R-squared, root-mean-square error (RMSE), mean absolute error (MAE), observation count, and raw-value variance. Constant candidates are rejected.

In the optimized full-library implementation, candidate ranking used the score:

$$\text{score} = R^2 - 0.10(\text{normalized RMSE}) + 0.05$$

The ten highest-ranked candidates were refitted using the stable shared regression implementation so that final scale, offset, RMSE, MAE, and R-squared values were calculated consistently. The output is therefore an interpretable signal definition rather than only a predicted numerical value.

3. Experimental Method

3.1 Synthetic CAN Generation

Synthetic traffic was used so that the true answer remained known for evaluation while discovery operated without access to it. Protocol databases were available only to the simulator and post-inference evaluator.

The key design choice was to generate independent random raw signal values rather than simulate realistic vehicle dynamics. This reduces accidental correlations between unrelated signals and makes mathematical identifiability easier to measure.

Raw integer values were generated first and physical values were then calculated using the true linear encoding. This prevents unnecessary quantization disagreement between the CAN payload and the synchronized known-value dataset.

3.2 Separation of Discovery and Ground Truth

The discovery algorithm receives only two clean data files. The first contains physical CAN frames with sample number, timestamp, message identifier, frame index, 29-bit CAN identifier, and raw payload bytes. The second contains synchronized physical measurements identified by opaque signal labels. The inference inputs intentionally omit PGNs, SPNs, bit locations, scales, offsets, byte order, signedness, and protocol database metadata.

Ground-truth JSON files, signal catalogs, Canboat definitions, and the J1939 database are available only to the evaluator after discovery. Automated leakage tests verify that the discovery modules can operate from the two clean input CSV files alone.

3.3 Candidate Search

Two principal search modes were developed. The aligned search considers byte-aligned signals with widths of 8, 16, and 32 bits. It tests signed and unsigned interpretations and applicable byte orders. The full generic search allows arbitrary start-bit positions and candidate widths of 1 through 32 bits, plus 40, 48, 56, and 64 bits. These candidate lengths are fixed algorithm parameters and are not obtained from the answer key.

3.4 J1939 Dataset

The complete J1939 experiment contained 2,043 message definitions and 13,774 signals. Of these, 6,930 were byte-aligned 8-, 16-, or 32-bit signals, 6,788 additional signals had widths of at most 32 bits, and 56 inferable fields were wider than 32 bits. No signals were classified as unsupported.

Population	Signals
Aligned 8/16/32	6,930
Additional <=32-bit	6,788
All widths 1-32	13,718
>32-bit inferable	56
Unsupported	0
Total	13,774

The traffic included 1,891 single-frame message definitions containing 12,165 signals, 149 BAM messages containing 1,575 signals, and three RTS/CTS definitions containing 34 signals. Long J1939 messages were transmitted using TP.CM and TP.DT frames and reassembled by the inference side without database metadata.

Transport	Message definitions	Signals
Single-frame	1,891	12,165
BAM	149	1,575
RTS/CTS	3	34

Total	2,043	13,774

The principal full-library experiment generated 200 observations per signal using deterministic random seed 42. This produced 2,754,800 target observations, 408,600 logical messages, and 547,600 physical CAN frames.

4. Results

4.1 Complete J1939 Recovery

Category/search	Signals	Top-1	Top-10	Raw full	Identifiable full
Aligned baseline	6,930	99.9278%	100.0000%	98.2107%	98.2828%
Extended <=32-bit	13,718	98.6587%	99.5626%	97.8787%	98.8118%
>32-bit	56	0.0000%	0.0000%	0.0000%	92.8571%
Complete library	13,774	98.2576%	99.1578%	97.4808%	98.7876%

Across the complete 13,774-signal database, Top-1 location accuracy was 98.2576%, Top-3 was 99.0417%, Top-5 was 99.0635%, and Top-10 was 99.1578%. Strict full-encoding accuracy was 97.4808%, while identifiability-adjusted full-encoding accuracy was 98.7876%.

Component	Accuracy / Error
Correct PGN	100.0000%
Correct start bit	99.6515%
Correct declared width	98.2576%
Correct byte order	100.0000%
Correct signedness	99.9419%
Mean absolute scale error	0.0756754
Mean absolute offset error	0.109431

The evaluator classified 13,427 signals as identifiable, 180 as structurally ambiguous, and 167 as incorrect. The 100% PGN recovery result is especially important because it shows that the algorithm always located the correct J1939 message even when it occasionally selected an observationally equivalent width or start-bit interpretation.

4.2 Effect of Search-Space Size

A more flexible search did not automatically improve Top-1 accuracy. The restricted aligned search required 967,480 candidate regressions and averaged approximately 140 candidates per target. The complete generic search evaluated 89,775,236 regressions and averaged approximately 6,518 candidates per target.

Search mode	Targets	Candidate regressions	Discovery runtime
Aligned	6,930	967,480	19.536 s
Full generic	13,774	89,775,236	83.069 s

The larger candidate space increased signal coverage but also introduced many more alternative explanations that could compete with the correct candidate. The complete end-to-end run, including simulation, both searches, evaluation, serialization, and plot creation, took approximately 142.76 seconds.

4.3 Signal Width

Small-width fields were generally recovered extremely accurately. Every 1-, 2-, 3-, 5-, 7-, 10-, 11-, 12-, and 21-bit signal in the tested database achieved 100% Top-1 accuracy. The 32-bit population was significantly more difficult: among 668 32-bit signals, Top-1 location accuracy was 93.2635%, and 151 of the 167 incorrect signals in the complete experiment were 32-bit fields.

The 56 signals wider than 32 bits produced zero strict Top-K location accuracy, but 92.8571% identifiability-adjusted full accuracy. Their valid physical ranges did not exercise the full declared width, leaving upper bits permanently zero. A shorter field starting at the same location could therefore reproduce the same physical mapping.

4.4 Transport Protocol Effects

Transport	Signals	Top-1	Identifiable full
Single-frame	12,165	98.8492%	99.7863%
BAM	1,575	93.7143%	91.1111%
RTS/CTS	34	97.0588%	97.0588%

The successful recovery of BAM and RTS/CTS signals demonstrates that the approach is not limited to individual eight-byte CAN frames. BAM was the most difficult transport subset and remains a useful target for further failure analysis.

5. Observation Count and Structural Ambiguity

A controlled J1939 experiment studied the effect of increasing the number of observations on a fixed 25-signal aligned population. Top-1 and Top-5 location accuracy were already 100% with ten observations. Strict full-encoding accuracy began at 88%, reached 92% at 200 observations, and remained at 92% even when the experiment was increased to 10,000 observations.

Observations	Top-1	Top-5	Full encoding	Runtime
10	100%	100%	88%	8.83 s
25	100%	100%	88%	9.52 s
50	100%	100%	88%	10.75 s
100	100%	100%	88%	13.18 s
200	100%	100%	92%	19.08 s
10,000	100%	100%	92%	61.99 s

The remaining two 10,000-observation failures had the correct PGN, start bit, width, byte order, factor, and offset. Only signedness differed. Because the valid unsigned raw values never entered the region where the sign bit was set, the signed and unsigned interpretations were observationally identical. This shows that additional data cannot resolve a property that never changes the observable output.

6. Identifiability

Identifiability became one of the most important findings of the project. A strict result requires PGN, start bit, bit length, byte order, signedness, scale, and offset all to match the database definition. The evaluator therefore distinguishes between a genuine inference error and a different encoding that is mathematically equivalent over the complete valid range.

Signedness is structurally ambiguous when every valid raw value is nonnegative and remains below the field's sign bit. In that case, signed and unsigned interpretations produce identical values. Width is similarly ambiguous when declared upper bits remain zero throughout the valid range and a shorter field at the same start position can represent the complete interval with the same scale and offset.

The project therefore reports both strict full-encoding accuracy and identifiability-adjusted accuracy. This preserves academic honesty: strict mismatches remain visible, while the adjusted metric does not incorrectly penalize the model for distinctions that are impossible to infer from the available information.

7. NMEA 2000 as an Earlier Validation Stage

Before J1939, the same general approach was developed against an NMEA 2000 simulator constructed from Canboat definitions. This stage established the architecture for raw traffic generation, synchronized known values, Fast Packet reconstruction, candidate extraction, least-squares regression, Top-K ranking, and strict separation between inference and ground truth.

A deterministic 25-target comparison later showed 96% Top-1 and Top-5 location accuracy and 96% strict full-encoding accuracy for the NMEA 2000 subset. The matched J1939 subset achieved 100% location accuracy, while strict full encoding ranged from 88% to 92%. These subset comparisons are not evidence that one protocol is inherently easier; instead, they show that the core mathematical inference architecture can be reused across different higher-level CAN protocols.

8. Computational Efficiency

Scaling the experiment to all 13,774 signals required reusing candidate extraction rather than processing the same raw candidate independently for every target. The optimized implementation groups logical traffic into streams, extracts candidate raw-value matrices in blocks, and regresses those candidates against multiple target signals. Target processing is divided into deterministic batches with checkpoint and resume support.

This optimization allowed the full generic experiment to perform 89,775,236 candidate regressions in approximately 83 seconds while keeping the peak candidate matrix near 2.44 MiB. The result shows that exhaustive-style candidate enumeration can remain computationally practical when extraction and regression are vectorized and reused.

9. Limitations

The primary experiments deliberately use independently randomized, noiseless signals. Real vehicles contain correlated quantities, sensor noise, calibration differences, timestamp jitter, packet loss, and latency. These effects may create stronger false correlations and reduce candidate separation.

The current inference model also assumes linear signal encoding. It is not designed to directly recover nonlinear lookup tables, encrypted fields, checksums, arbitrary binary blobs, or state-machine behavior.

Some database definitions are fundamentally not identifiable from numerical observations. Declared widths and signedness can remain impossible to distinguish even with infinitely many observations if the valid signal range never exercises the relevant bits. Finally, 32-bit and BAM signals account for a disproportionate share of the remaining incorrect results and deserve more detailed analysis.

10. Future Work

Future experiments should introduce controlled measurement noise and study recovery accuracy as a function of both noise level and observation count. A second extension should introduce controlled physical

correlation between signals, such as pressure partially depending on engine speed, to determine whether the ranking system can distinguish the true encoding from a correlated but incorrect field.

The remaining 167 incorrect J1939 signals should also be categorized by failure mechanism. Particular attention should be given to the 151 incorrect 32-bit signals and the reduced BAM performance. Another useful extension would be to report equivalence classes of candidate encodings when multiple definitions are mathematically indistinguishable over the entire valid signal range.

11. Conclusion

This project demonstrates that a simple and interpretable mathematical method can recover most of a large CAN signal database without access to that database during discovery. Using only raw CAN traffic and synchronized known physical values, the system searches candidate bit fields and estimates their scaling using least-squares regression.

On the complete supplied J1939 database, the system analyzed all 13,774 signals, correctly identified the PGN for every signal, achieved 98.2576% Top-1 location accuracy, 99.1578% Top-10 location accuracy, 97.4808% strict full-encoding accuracy, and 98.7876% identifiability-adjusted full-encoding accuracy.

The results also show that the final few percentage points cannot be understood purely as conventional model error. Some properties of a CAN definition are not observable from the available values. If signed and unsigned interpretations are identical throughout the valid range, or if the upper bits of a wide field can never become nonzero, the database's declared representation cannot be uniquely reconstructed from observations alone.

The central result is therefore broader than a high decoding accuracy. The experiment demonstrates both what can be learned from synchronized CAN data and where the mathematical limits of that learning lie. Under the controlled linear conditions tested here, systematic candidate search and linear regression are sufficient to reconstruct the overwhelming majority of a complete J1939 signal library, including transported messages, without giving the discovery algorithm access to the original signal map.

References

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All reported performance values in this paper are measured results from the completed project experiments.